RESTAURANT ANALYTICS
Business Intelligence Report

A Data-Driven Analysis of Customer Behaviour, Sales Performance and Delivery Operations

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Programme	BS Software Engineering & Artificial Intelligence
Dataset	clean_data.csv — 6,000 restaurant orders, Aug 2023 – Aug 2025
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1. Executive Summary

This report analyses 6,000 restaurant orders placed through a food-ordering platform operating across five Pakistani cities — Lahore, Karachi, Islamabad, Multan and Peshawar — between August 2023 and August 2025. The dataset covers orders from five restaurant partners (McDonald's, KFC, Pizza Hut, Subway and Burger King) and captures customer demographics, order details, payment behaviour, loyalty activity and delivery outcomes.

The platform generated a total of **PKR 14.34 million** in recorded revenue across the two-year window, at an average order value of **PKR 2,391**. Demand is spread fairly evenly across cities, restaurant partners and cuisine categories, with no single segment dominating the business — the largest gap between any two cities is under 16%. The most significant finding in the dataset is operational rather than commercial: only **34.3%** of orders were marked as successfully delivered, while the remaining **65.7%** were either delayed or cancelled. This delivery reliability gap is far more consequential to the business than any revenue trend uncovered in the data.

Customer-level metrics such as loyalty points and order frequency showed almost no relationship with either revenue or churn in this dataset, and an attempt to build a predictive churn model using the available fields performed no better than random guessing (AUC ≈ 0.50). Rather than treat this as a dead end, the report treats it as a genuine finding in its own right: the dataset appears to capture a single transaction per customer rather than a full purchase history, which limits how far behavioural prediction can go without richer, longitudinal data. Section 8 discusses this in more depth and what it implies for future data collection.

Overall, the analysis points to three priority areas for the business: fixing delivery execution (the single biggest lever on customer experience), reducing continued reliance on cash payments in favour of digital wallets and cards, and redesigning the loyalty programme so that it is actually tied to repeat engagement rather than functioning as a static points counter.

2. Project Overview

The purpose of this project is to apply data analysis and business intelligence techniques to a real-world style restaurant ordering dataset, in order to understand how customers interact with a food delivery platform and where the business can improve. This was carried out as an academic exercise for the Software Engineering & AI programme, using Python as the primary analysis environment in place of SPSS, alongside standard data science libraries (pandas, matplotlib and scikit-learn).

The project set out to answer a specific set of business questions:

- • Which cities, restaurant partners, cuisines and dishes generate the most revenue, and how concentrated is that demand?
- • How do customers prefer to pay, and does payment method relate to order value or delivery outcome?
- • How reliable is order fulfilment, and does that reliability vary by restaurant, city or payment method?
- • Is there a meaningful relationship between loyalty points, order frequency, ratings and customer churn?
- • Can the available fields support predictive modelling (e.g. churn prediction), and if not, why not?

The scope of the analysis is deliberately limited to what the dataset can actually support. Where the data does not contain a strong signal — for example, around churn drivers — the report says so explicitly rather than manufacturing a narrative around it. This is intended to reflect the standard a working analyst would be held to: recommendations are only as good as the evidence behind them.

3. Dataset Description

The dataset (**clean_data.csv**) contains **6,000 records** across **20 columns**, with each row representing one customer order. Every column was populated — there were no missing values — and the schema splits naturally into four groups of information:

3.1 Column Groups

Group	Fields
Customer profile	customer_id, gender, age group, city, signup_date, loyalty_points, churned
Order details	order_id, order_date, restaurant_name, dish_name, category, quantity, price, payment_method
Engagement history	order_frequency, last_order_date
Post-order feedback	rating, rating_date, delivery_status

3.2 A Structural Note on the Data

One detail worth flagging before any analysis: the dataset contains exactly **6,000 unique customer IDs and 6,000 unique order IDs** — a perfect one-to-one match. In other words, every customer in this file appears with exactly one order. This means the dataset is best understood as a **single-transaction snapshot across 6,000 different customers**, rather than a purchase history that follows the same customers over multiple orders. Fields like *order_frequency* and *loyalty_points* are therefore pre-existing account-level attributes carried over from a wider customer profile, not figures that can be recalculated from the transactions in this file. This distinction matters a great deal for how far the data can be pushed analytically, and it is referenced again in Sections 7 and 8.

3.3 Coverage

- Time period: 23 August 2023 to 22 August 2025 (24 full months, plus a partial final month).
- Cities: Lahore, Karachi, Islamabad, Multan, Peshawar.
- Restaurant partners: McDonald's, KFC, Pizza Hut, Subway, Burger King.
- Cuisine categories: Fast Food, Italian, Chinese, Continental, Dessert.
- Order value range: unit prices from PKR 100 to PKR 1,500; quantities from 1 to 5 items per order.

4. Data Cleaning and Preprocessing

Before any analysis, the dataset was checked for the usual data-quality issues — missing values, duplicate records, inconsistent categories and incorrectly typed columns. The file turned out to be genuinely clean going in, which kept preprocessing to a short, well-defined set of steps rather than an extensive repair job:

- • **Missing values:** none were found across any of the 20 columns — confirmed with a full null-count check.
- • **Duplicates:** no duplicate customer_id or order_id values were present.
- • **Date parsing:** the four date columns (signup_date, order_date, last_order_date, rating_date) were stored as text and converted to proper datetime objects to allow time-based analysis (monthly trends, day-of-week patterns).
- • **Category consistency:** categorical fields (city, restaurant_name, category, payment_method, delivery_status, churned, gender, age) were checked for spelling variants or stray whitespace — none were found; each field held a small, clean set of consistent labels.
- • **Derived field:** a *revenue* column was created as quantity × price, since order value was not supplied directly and is needed for almost every downstream calculation.
- • **Structural check:** customer and order counts were cross-verified (6,000 vs 6,000) to confirm the one-order-per-customer structure described in Section 3.2, since this affects how metrics like order_frequency should be interpreted.

No rows were removed and no values were imputed, since the dataset did not require it. This is worth stating plainly rather than describing an elaborate cleaning process that did not actually take place — the raw file was already analysis-ready.

5. Exploratory Data Analysis

This section walks through how orders, revenue and customer behaviour are distributed across the main dimensions in the dataset: time, geography, restaurant partner, cuisine, payment method, delivery outcome and customer rating.

5.1 Revenue Trend Over Time

Monthly revenue peaked early, in September 2023 (PKR 696,622), shortly after the observed data begins, and has drifted gradually downward since. Comparing the first six full months of data to the most recent six full months, average monthly revenue fell from roughly PKR 623,000 to PKR 565,000 — a decline of about 9.3%. The trend is not a sharp collapse; it looks more like a slow, steady softening layered on top of a lot of month-to-month noise.

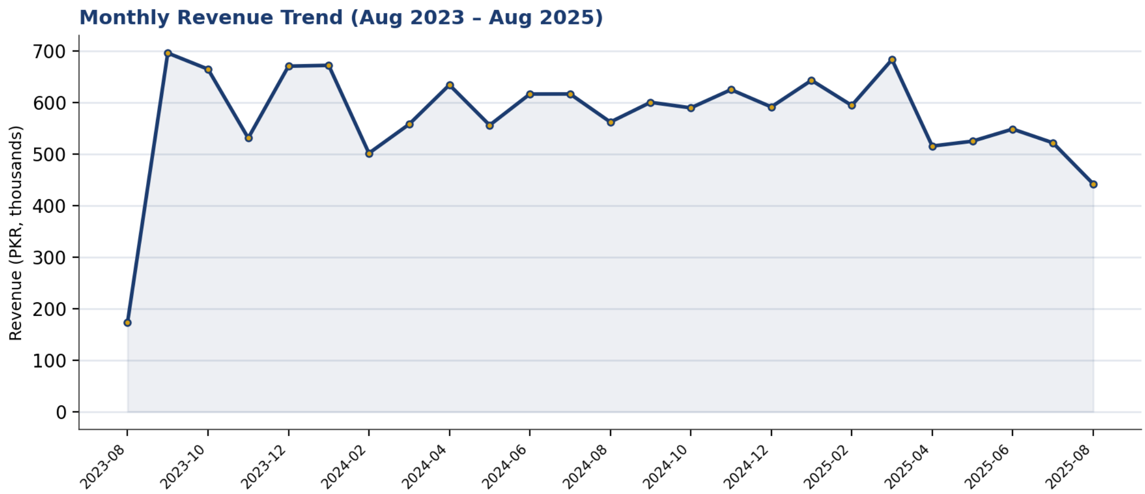
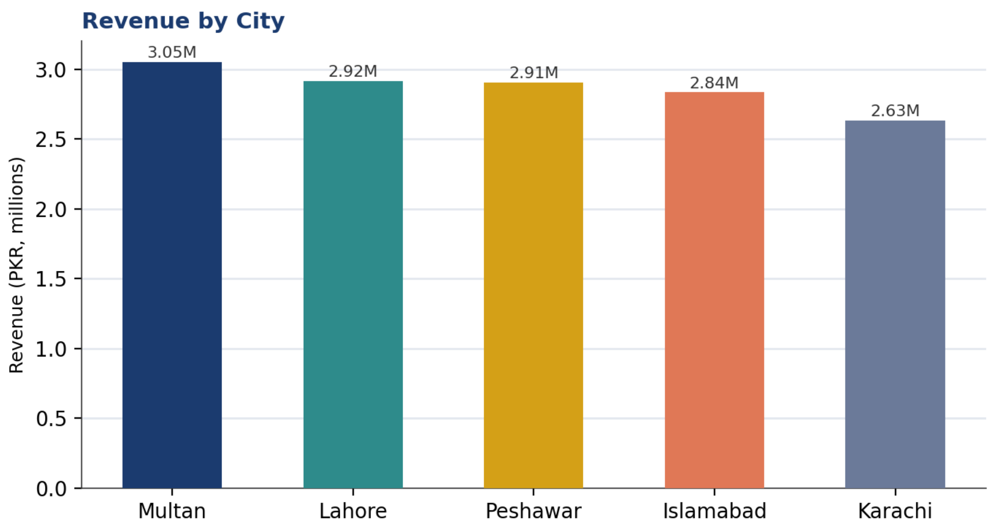


Figure 1. Monthly revenue, August 2023 – August 2025.

5.2 Revenue by City

Revenue is spread fairly evenly across the five cities, which suggests demand is not concentrated in one location. Multan leads at PKR 3.05 million, narrowly ahead of Lahore and Peshawar, while Karachi sits lowest at PKR 2.63 million — a gap of just under 16% between the top and bottom city. For a platform operating in five different metro markets, this is a reasonably balanced footprint.



5.3 Revenue by Restaurant Partner and Cuisine

Among the five restaurant partners, Pizza Hut generated the most revenue (PKR 3.00 million) and McDonald's the least (PKR 2.64 million), with the other three partners clustered in between. At the cuisine level, Italian dishes (largely driven by Pizza Hut and Subway's pasta and pizza items) led with PKR 3.01 million, while Dessert trailed at PKR 2.65 million. As with the city breakdown, no single partner or cuisine dominates — the business is not overly dependent on any one relationship.

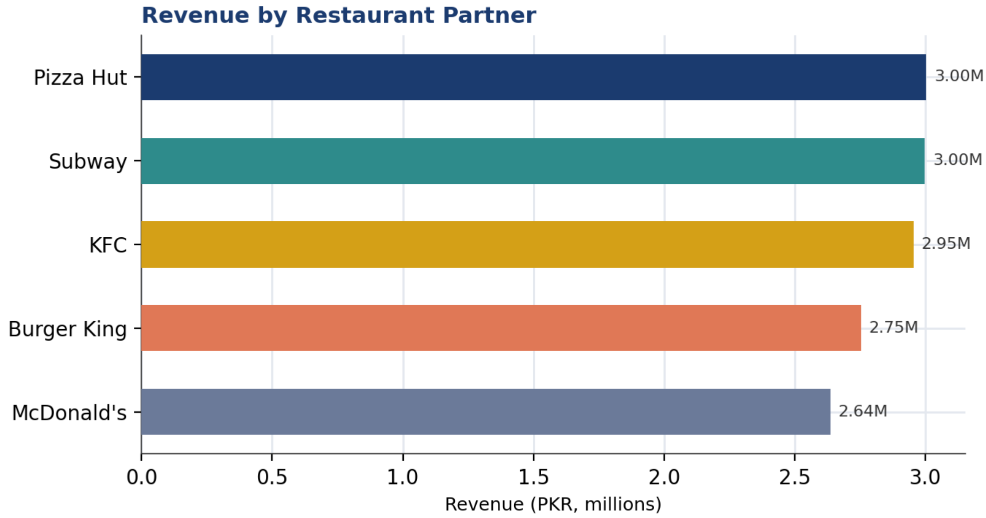


Figure 3. Total revenue by restaurant partner.

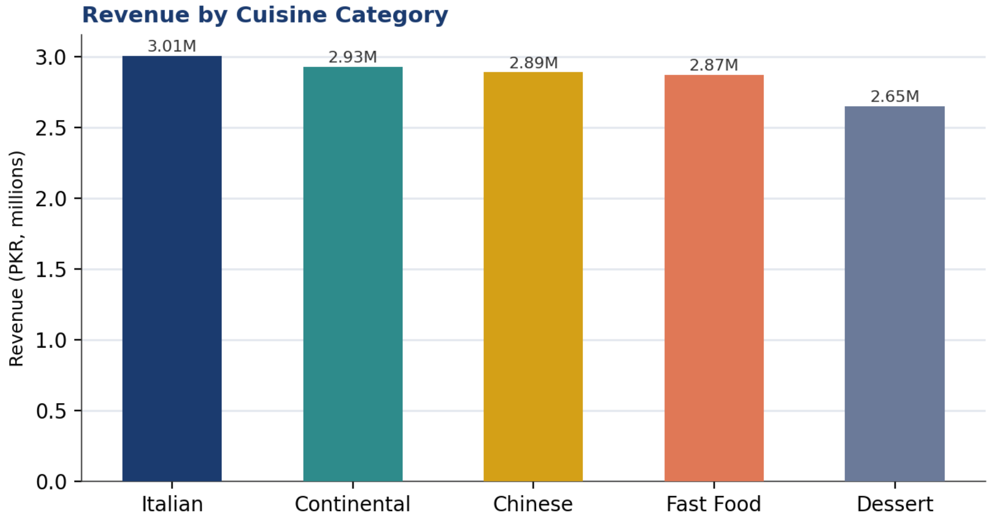


Figure 4. Total revenue by cuisine category.

5.4 Payment Methods

Cash remains the most common way customers settle their orders, accounting for 34.6% of total revenue, ahead of Wallet (32.9%) and Card (32.5%). The gap between the three methods is narrow, but cash's continued lead is notable for a digital ordering platform — it implies a meaningful share of the customer base is still paying on delivery rather than settling in-app.

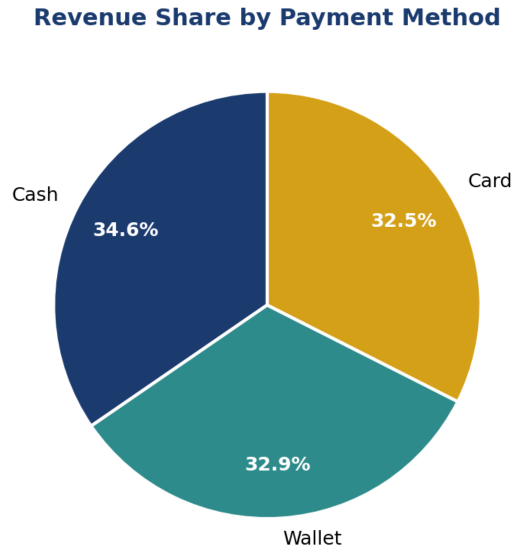


Figure 5. Revenue share by payment method.

5.5 Delivery Outcomes

This is the sharpest pattern in the entire dataset. Only 34.3% of orders were marked **Delivered**; the rest were split almost evenly between **Delayed** (32.9%) and **Cancelled** (32.8%). Put differently, roughly two out of every three orders placed on the platform did not complete normally. This is a substantially bigger issue than any of the revenue or demographic patterns discussed elsewhere in this report, and it is examined further in Sections 7 and 8.

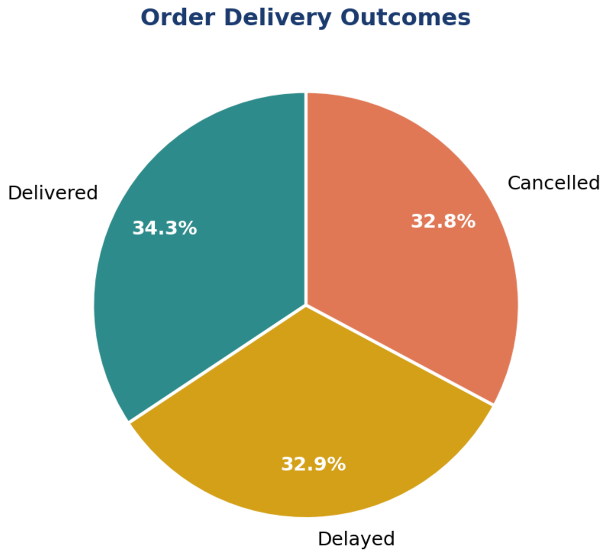


Figure 6. Distribution of order delivery outcomes.

5.6 Ratings

Customer ratings average 3.0 out of 5 and are almost evenly spread across all five star values (each rating from 1 to 5 accounts for roughly 19–21% of orders). There is no skew toward very satisfied or very dissatisfied customers — sentiment sits squarely in the middle, which is consistent with a delivery experience that is inconsistent rather than uniformly good or bad.

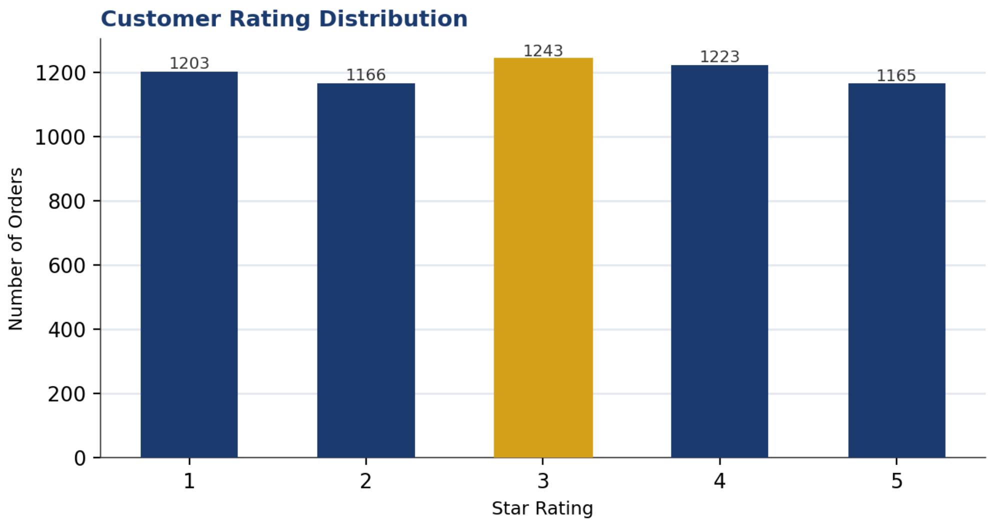


Figure 7. Distribution of customer ratings.

5.7 Day-of-Week Patterns

Thursday is the strongest day for revenue (PKR 2.17 million), while Tuesday is the weakest (PKR 1.91 million) — a difference of about 14%. Weekend revenue (Friday through Sunday) is solid but not dramatically higher than weekday revenue, suggesting demand for this platform is not purely a weekend, leisure-driven habit.

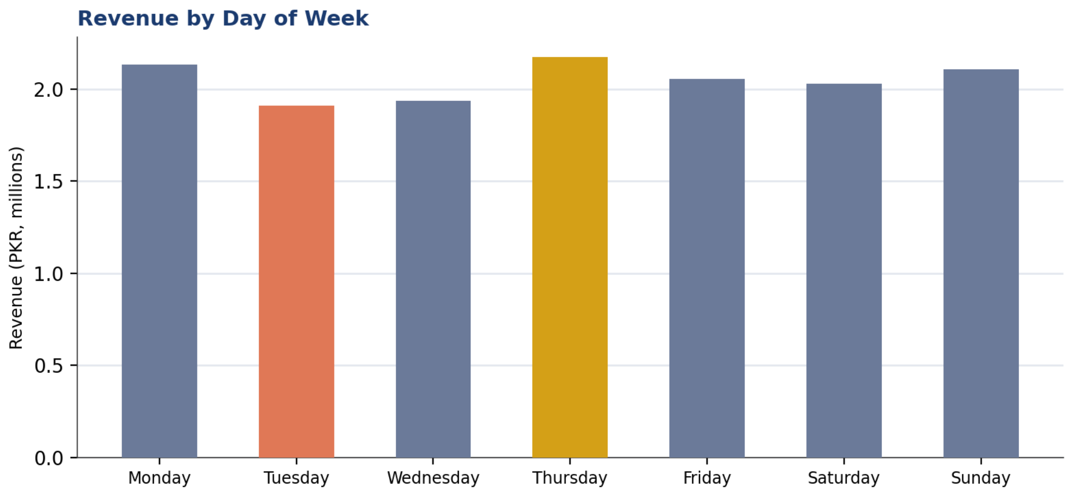


Figure 8. Revenue by day of week.

6. Key Performance Indicators

The table below summarises the core metrics used to track platform performance across the full two-year window.

Metric	Value
Total Revenue	PKR 14,343,912
Total Orders	6,000
Unique Customers	6,000
Average Order Value	PKR 2,390.65
Average Items per Order	2.99
Average Rating	3.00 / 5
On-Time Delivery Rate	34.3%
Delayed / Cancelled Rate	65.7%
Customer Churn Rate	49.7%
Average Order Frequency (account-level)	25.3 orders
Average Loyalty Points Balance	250
Cash Payment Share	34.6% of revenue

Two figures stand out against the rest: the **65.7% delayed/cancelled rate** and the **49.7% churn rate**. Both point to the same underlying story — roughly half the customer base is not sticking with the platform, and the fulfilment experience gives them good reason not to.

7. Business Insights

7.1 Delivery Execution Is the Platform's Core Problem

With only 34.3% of orders delivered successfully, delivery reliability is by far the most urgent issue in this dataset. Breaking this down by restaurant partner shows some separation: KFC completes 37.3% of its orders successfully and Burger King the fewest, at 31.8%. The gap between partners (roughly 5.5 percentage points) is modest relative to the roughly 1,150–1,260 orders behind each restaurant's figures, so it should be treated as a lead worth investigating operationally rather than a settled conclusion — but it is consistent with the broader pattern of uneven fulfilment across the board.

7.2 Cash Is Still the Default Payment Method

Despite operating as a digital ordering platform, cash accounts for the largest share of revenue (34.6%), narrowly ahead of digital wallets and cards. This has practical implications: cash-on-delivery orders are typically harder to reconcile, slower to process at the door, and offer the platform less visibility into failed or disputed transactions compared with pre-paid digital methods.

7.3 Demand Is Broad-Based, Not Concentrated

No single city, restaurant partner, or cuisine category accounts for a disproportionate share of revenue — the largest spread across any of these dimensions is under 16%. This is a healthy sign in one respect (the business is not overexposed to any one market or partner) but it also means there is no obvious "star performer" segment to double down on. Growth is more likely to come from lifting reliability and conversion across the board than from concentrating investment in a single city or partner.

7.4 Loyalty Points Are Not Currently Driving Behaviour

Loyalty point balances show essentially no relationship with order value (correlation of -0.004) or with order frequency (correlation of 0.007) in this dataset. Customers who churned actually held marginally *more* loyalty points on average (252) than active customers (248). At face value, this suggests the loyalty programme functions as a passive counter rather than an active retention lever — accumulating points does not appear to be translating into repeat engagement.

7.5 Larger Orders Carry Disproportionate Revenue Weight

Orders of four or five items make up only 39.2% of total order volume but contribute 59.2% of total revenue. This is a meaningful, if unsurprising, pattern: a comparatively small share of larger, bulk-style orders is doing most of the revenue heavy lifting, which makes upselling and bundling strategies (discussed in Section 9) a natural place to focus.

7.6 Weekday Timing Matters, but Modestly

Thursday generates about 14% more revenue than the slowest day, Tuesday. This is a real but fairly gentle pattern — not the kind of dramatic weekend spike often assumed in restaurant delivery businesses. It is still useful for staffing and promotional timing, particularly around the Tuesday trough.

8. AI-Driven Insights

Beyond descriptive analysis, two machine learning approaches were tested on this dataset: a churn prediction model and a customer segmentation exercise. The results of both are reported honestly below, including where the models did not perform well — because understanding *why* a model fails is often as useful to a business as a model that succeeds.

8.1 Churn Prediction: A Negative but Useful Result

A random forest classifier was trained to predict customer churn using demographics, order details, payment method, loyalty points, order frequency and rating. On a held-out test set, the model achieved an AUC of only 0.495 — statistically indistinguishable from random guessing (0.50 = no predictive power). A separate model built to predict star rating from the same fields performed even worse, with a negative R^2, meaning a simple average would have predicted ratings better than the model did.

This is not a failure of modelling technique; it is a direct consequence of the data structure identified in Section 3.2. Because each customer appears with only one order, the dataset cannot show how an individual customer's behaviour changes over time — which is normally the strongest signal in any churn model (declining order frequency, worsening ratings, growing gaps between orders). Fields like loyalty_points and order_frequency are static snapshots rather than trends, so a model has little to learn from beyond the one transaction on record. The genuine AI-driven insight here is therefore a data-collection one: to build a working churn model, the business needs to link full order histories per customer over time — not just a single transaction snapshot with pre-aggregated account fields.

8.2 Customer Segmentation (K-Means Clustering)

A three-cluster K-means segmentation was run on order value, loyalty points and order frequency to see whether natural customer groupings exist. Three distinct segments emerged:

Segment	Customers	Avg. Order Value	Avg. Loyalty Pts	Avg. Frequency	Description
High-Value Orders	1,373	PKR 5,047	254	26.3	Large, high-revenue single orders — the most valuable transa
Loyalty-Rich, Low-Spend	2,241	PKR 1,590	382	25.7	Highest loyalty balances but comparatively small order value –
Low-Engagement	2,386	PKR 1,614	124	24.3	Lowest loyalty balance and smallest orders — likely newer or

The most actionable finding here is the middle segment: customers holding the highest loyalty balances (382 points on average) are not the platform's biggest spenders — their average order value (PKR 1,590) is the lowest of the three groups. This reinforces the point made in Section 7.4: the loyalty programme, as currently structured, has accumulated engagement without converting it into higher-value orders.

8.3 Delivery Risk by Restaurant Partner

Modelling delivery outcome as a function of restaurant, city and payment method did not surface a dominant single driver, but restaurant partner showed the clearest (if moderate) separation: KFC and Pizza Hut complete orders successfully somewhat more often than Burger King and Subway. A production version of this model would benefit from operational data not present here — rider assignment, distance from kitchen to customer, and order preparation time — which would likely explain far more of the variation in delivery outcomes than customer-side attributes do.

9. Actionable Business Recommendations

1. Treat delivery reliability as the top priority

With 65.7% of orders delayed or cancelled, this is the single largest lever on customer experience and, by extension, revenue and churn. Start by auditing the operational causes behind cancellations at Burger King and Subway specifically, since they show the weakest completion rates, before rolling any fix out platform-wide.

2. Push customers toward digital payment methods

Cash still accounts for over a third of revenue. Incentivising wallet or card payment — through a small discount, faster checkout, or guaranteed refund handling for digital payments — would reduce reconciliation overhead and give the business better visibility into failed transactions.

3. Redesign the loyalty programme around real incentives

Loyalty points currently show no relationship with order value or repeat engagement. Consider restructuring the programme around tiered, expiring rewards (e.g. a discount unlocked after a points threshold within a time window) rather than an open-ended balance that customers can accumulate without acting on.

4. Use bundling and upsell prompts around the 4–5 item order size

Since orders of four or more items generate 59% of revenue from only 39% of order volume, prompting customers toward a fourth or fifth item at checkout (e.g. a suggested side or dessert) is a low-cost way to shift more orders into this higher-value band.

5. Time promotions around the Tuesday demand trough

Tuesday is consistently the softest day for revenue, roughly 14% below Thursday, the strongest day. A targeted Tuesday discount or bundle could help smooth demand and make better use of kitchen and rider capacity that would otherwise sit underused.

6. Invest in longitudinal customer data before building churn models

The current dataset captures one order per customer, which is why churn and rating prediction both performed at chance level in this analysis. Before investing further in predictive modelling, the platform should prioritise linking full order histories to individual customers over time — this single change would likely do more for future analytics than any additional feature engineering on the current data.

10. Conclusion

This analysis of 6,000 restaurant orders shows a platform with a healthy, broad-based demand base — revenue is spread reasonably evenly across cities, restaurant partners and cuisines, and average order values are consistent across customer segments. The commercial fundamentals are not the concern here.

The concern is operational. Nearly two-thirds of orders did not complete as delivered, and roughly half the customer base is currently inactive. These two figures are almost certainly connected, and fixing delivery execution is likely to move both revenue and retention more than any pricing, marketing or loyalty change attempted in isolation.

On the analytics side, the attempt to build predictive churn and rating models produced a genuinely useful negative result: with only one order per customer on record, the data does not yet contain the kind of longitudinal signal that predictive models need. This is a data-collection gap rather than a modelling failure, and closing it — by linking repeat orders to the same customer over time — should be treated as a prerequisite for any future churn or personalisation work, rather than something to be solved through more sophisticated algorithms applied to the current data.

Taken together, the recommendations in Section 9 are ordered roughly by expected impact: delivery reliability first, followed by payment method shifts, loyalty programme redesign, and order-size upselling, with better data collection as the foundation that will make future analysis — including AI-driven churn prediction — actually possible.